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Information Integration as a Basic Cognitive Process

RAMADHAR SINGH*
Indian Institute of Management Bangalore

ABSTRACT

When Indians are provided with multiple pieces of information about a situation and asked to make a judgment, how do they do it? This chapter documents that Indians assign values and importance to different pieces of given information and integrate them by intuitive rules of adding, subtracting, multiplying, dividing, or averaging. These processes – collectively called cognitive algebra – reflect casual beliefs about how different pieces of information influence each other. Considerable evidence for cognitive algebra n judgements of social groups, leadership situations, job satisfaction, outcome allocation, disciplinary action, and predictions of performance and gift size shows that information integration is a basic cognitive process. Further, the cognitive algebra is useful in identifying the: (a) cultural and developmental differences in judgements; and (b) imputations, that is, inferences that people make about missing information from given information. Overall, Indians have high cognitive capacity.

BACKGROUND

Suppose you have been offered a new job that pays *high salary* and *hefty bonus* but requires you to do *uninteresting tasks* and *too much travelling,* how attracted would you be to this job offer? Should you accept the offer, would you be satisfied with the job? Both judgements require you to integrate four pieces of information. How would you integrate them?

When my collaborators and I proved Indians with multiple pieces of information about a target and asked them to judge it, the judgements they rendered were reflective of intuitive adding, subtracting, multiplying, dividing, averaging, or a combination of them – collectively called 'cognitive algebra' (Anderson 1974). Interesting in cultural difference was much less when we began our work that in recent years (Lehman *et al*., 2004).

- This chapter is dedicated to Norman H. Anderson for his extraordinary roles to the author's development as a person as well as a scientist.

As our research progressed, however, we became interested in whether the findings were unique to Indians or not. So, this chapter documents that information integration is a basic cognitive process, and that cognitive algebra is an excellent criterion for cultural and developmental comparisons.

If cognitive algebra is a basic process, it should be present in judgements by children. Accordingly, Singh *et al.* (1978) provided 6–7-year-olds with information about the number of toys available to children and the ratio of good to bad members in sixteen playgroups, and obtained judgements of playgroup attractiveness. The response measure was a 9-point rating scale, anchored by 1 (lowest) and 9 (highest). Figure 5.1 presents mean attraction towards playgroups as function of the number of toys and the ratio of good to bad members in the groups.

If children integrated the information given by an adding rule, then the four curves of Figure 5.1 should be parallel, that is, the differences between the curves should be constant. The curves of Figure 5.1 have the pattern of *parallelism*. The quantitative confirmation of the pattern of parallelism came from a statistically non-significant

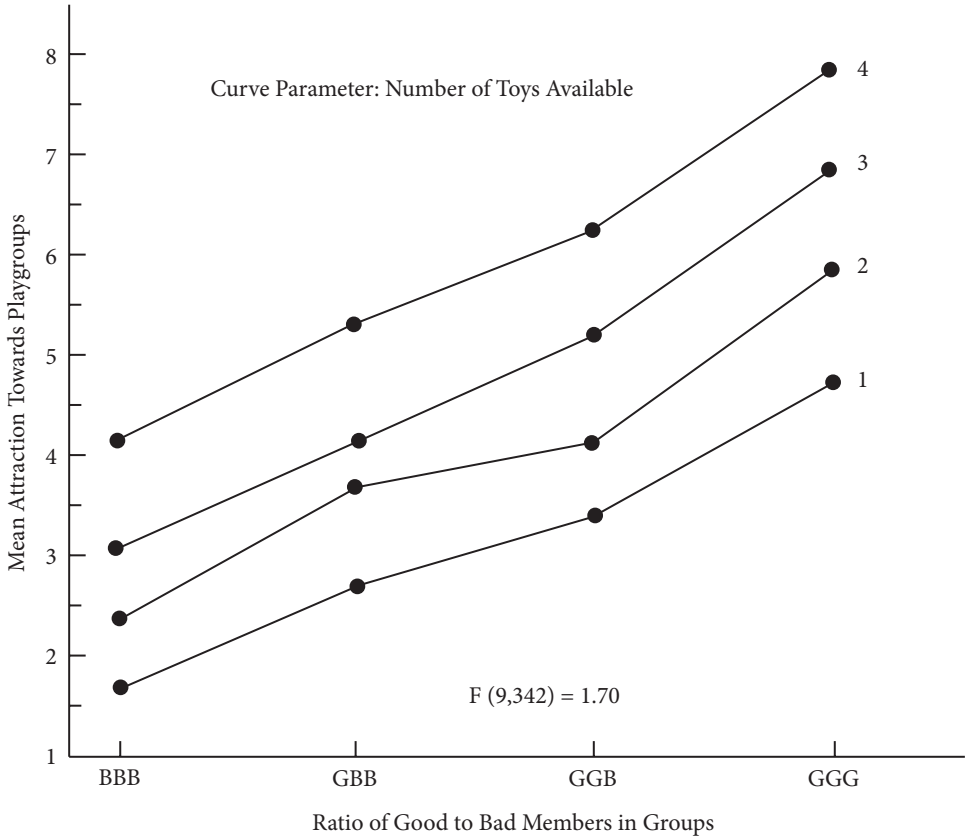

Figure 5.1 Mean Attraction towards Playgroups as a Function of the Number of Toys Available (curve parameter) and the Number of Good (G) and Bad (B) Members (listed on the horizontal axis)

Source: Singh *et al.* (1978). Reproduced with permission from Elsevier.

interaction effect in analysis of variance (ANOVA) (Anderson 1981, 1982). Thus, results supported an additive model:

Playgroup attractiveness = Number of toys + Ratio of good to bad members. (1)

Given the foregoing evidence for an additive model, we estimated the subjective values assigned by children to the information given. These values are the marginal means in ANOVA. The attractiveness values of 1, 2, 3, and 4 toys were 3.12, 3.98, 4.81, and 5.88, respectively, which differed about equally in proportion to the number of toys. Similar attractiveness values of the four ratios of good to bad members in playgroups were 2.83, 3.96, 4.73, and 6.29. The first three means are equally spaced; the fourth one is out of line. Because the first three groups consisted of at least one bad member but the fourth group had all four good members, this difference in attractiveness values denoted a stronger effect of bad than good members.

Information Integration Theory and Functional Measurement

The experiment on playgroup attractiveness was guided by the methods of Anderson's (1981, 1982) information integration theory and functional measurement. The theory deals with how people integrate different pieces of information in a judgement. Mounting evidence shows that diverse kinds of judgements obey cognitive algebra, and that the '…integration function implicitly contains the measurement scales for stimuli and response' (Anderson 1981: 4). This interdependence between cognitive algebra and functional measurement is the foundation of information integration theory.

To determine cognitive algebra, seemingly different pieces of information are presented in a Row × Column factorial design.[1] The experiment is conducted, and the results are plotted as a function of the information manipulated. The pattern in the factorial plot indicates which integration rule operated. Figure 5.1 illustrates a 4 × 4 factorial design; the pattern of parallelism in it reveals that the two kinds of information were added together; and the estimated attractiveness values of the two factors indicate how the adding rule yielded those functional values.

When both the values of the information given and the integration rule used are unknown, how can the pattern in the factorial plot of the results diagnose the cognitive algebra underlying the judgements? Because the integration rule is the key in functional measurement, the known rules of adding, averaging, or multiplying serve as the starting point. Simply assign some values to the different pieces of information, use one of these rules with those numbers, and generate the predicted pattern of judgements. When the pattern in the factorial plot of the judgements conforms to the pattern generated by the known rule, then the underlying cognitive algebra is adjudged to be as such.

Adding, Averaging, and Multiplying Rules

Consider, for instance, a hypothetical experiment on judgement of suitability of sites offered by different states to a multinational corporation for starting a new venture in India. Suitability of a site depends upon the ease of the land acquisition (Row Factor A) and the uninterrupted supply of electricity (Column Factor B). Both factors can be realistically manipulated by suitable phrases representing the low (L), moderate (M), and high (H) levels and, hence, would constitute a 3 × 3 (Row × Column) design. Assign arbitrarily ordinal values of 3.5, 5, and 6.5 to the respective L, M, and H levels of the row factor

and of 1, 5, and 9 to the corresponding three levels of the column factor, and take their sums, averages, and products. The upper three rows of Table 5.1 list nine values from each of the three known rules of adding, averaging, and multiplying; Figure 5.2 plots them graphically from left to right.

The three curves from both the adding and averaging rules in Figure 5.2 constitute the pattern of parallelism. The three curves from the multiplying rule in the rightmost graph of Figure 5.2, by contrast, form a *linear fan pattern*. The effect of the row factor is more when the value of the column factor is high than when it is low and vice versa. This non-additive pattern implies an interaction effect in ANOVA.

Adding versus Averaging

In generating the three patterns in the solid curves of Figure 5.2, we exclusively relied on the subjective values assigned to the three levels of the row and column factors. Both the adding and averaging rules engender the same pattern of parallelism. How can the adding rule be then distinguished from the averaging rule?

Add a moderate-valued positive piece of information about the row factor to a high- or a low-valued positive piece of information about the column factor. The moderate-valued positive piece of information about the row factor should increase the value of both the high- and low-valued piece of information about the column factor if *added* but should *average down* the high-valued positive information about the column factor. This opposite effect of the addition of a moderate-valued positive piece of information to high-valued positive information follows from the averaging model (Anderson 1965).

Now, look again at the precise values assigned to levels of the row and column factors in Table 5.1. The judgement from the H level of the column factor alone is higher than that from the H levels of both the row and column factors. The judgement from the L level of the column factor alone is lower than that from the L levels of both the row and column factors. Why? The average of the moderate-valued positive information about the row factor and the extremely positive information about the column factor is lower than the average of the extremely positive information about the column factor alone. Likewise, the average of the moderate-valued negative information about the row factor and the extremely negative information about the

Table 5.1 Adding, Averaging, and Multiplying Rules Illustrated Numerically for an Experiment

Levels and Values of the Row Factor A: Ease of Land Acquisition		Levels and Values of the Column Factor B: Supply of Electricity								
		Adding Rule			Averaging Rule			Multiplying Rule		
		L	M	H	L	M	H	L	M	H
Levels	Values	1	5	9	1	5	9	1	5	9
H	6.5	7.5	11.5	15.5	3.75	5.75	7.75	6.5	32.5	58.5
M	5.0	6	10	14	3	5	7	5	30	45
L	3.5	4.5	8.5	12.5	2.25	4.25	6.25	3.5	17.5	31.5
Unknown		1	5	9	1	5	9	1	5	9

Source: Author.

Note: The nine values are the sums, averages, and products of the arbitrary values assigned to the low (L), moderate (M), and high (H) levels of the row and column factors. The values in the bottom are the levels of the Row Factor of B alone.

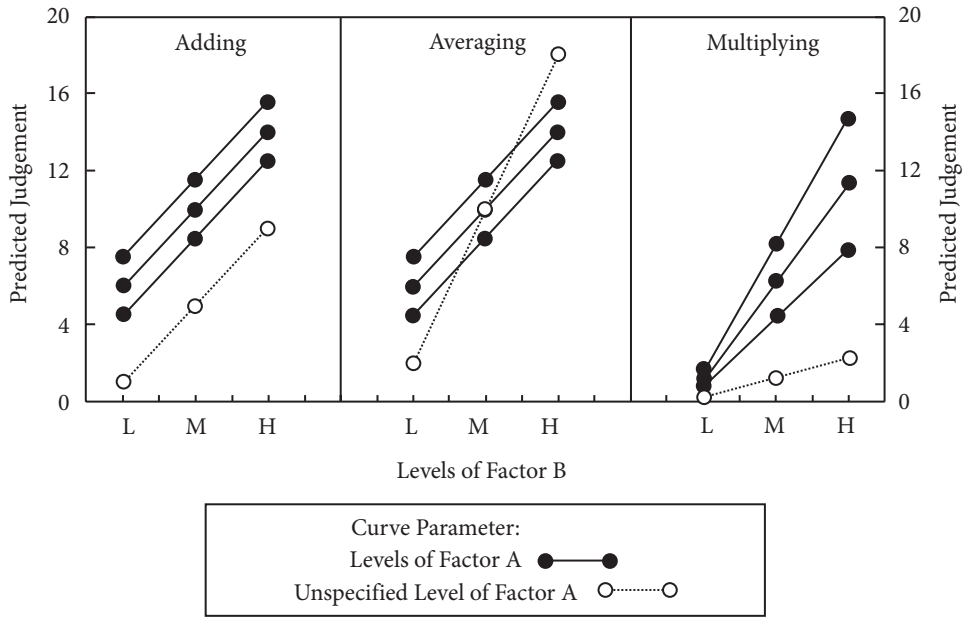

Figure 5.2 The Row × Column Factorial Plots of Judgements Predicted by the Adding, Averaging, and Multiplying Rules in Table 5.1

Source: Author.
Notes: (a) The dashed curve is based on information about Factor B alone.
(b) L, Low; M, Moderate; and H, High.
(c) The numerical values from the multiplying rule, listed in Table 5.1, were divided by 4 and those from the averaging rule were multiplied by 2, to accommodate all the three graphs with a uniform ordinate and make the patterns predicted by the three rules stand out in a single figure.

column factor is higher than the average of the extremely negative information about the column factor alone. These opposite effects of the moderate-valued information added to the high- and low-valued information produce the *crossover pattern* in the joint plot of the solid and dashed curves in Figure 5.2 and, hence, distinguish the averaging rule from the adding one.

Evidence for Cognitive Algebra

Averaging Model

Playgroup Attractiveness

To distinguish between the averaging and adding rules for the additive effect in Figure 5.1, Singh *et al.* (1978) paired two slightly good (G^-G^-) or two slightly bad (B^-B^-) members with playgroups already consisting of only two good (GG) or only two bad (BB) members. As predicted by the averaging rule, mean attraction was higher to a GG playgroup (M = 7.88) than to a GGG^-G^- playgroup (M = 6.61), but mean attraction was lower to a BB playgroup (M = 2.17) than to a BBB^-B^- playgroup (M = 4.33).

Situational Favourableness for Leaders

In Fiedler's (1967) contingency theory, one determinant of leadership effectiveness is *situational favourableness*: '… the degree to which the situation provides the leaders with a potential power and influence over the

group behaviour' (Fiedler 1971: 129). Situational favourableness depends upon group atmosphere (GA), task structure (TS), and position power (PP). Factorial combinations of the low versus high levels of these three factors generate eight levels of situational favourableness for the leader. Consistent with rank orders of those eight situations vis-à-vis favourableness for their respective leaders, Nebeker (1975) proposed an adding model to account for his results.

There was an apparent inconsistency between the adding conclusion by Nebeker (ibid.) and Singh *et al.*'s (1978) finding showing that children integrated social group information by averaging. Thus, Singh, Bohra *et al.* (1979) tested the rule underlying the judgements of situational favourableness for leaders. Leadership situations were generated from the low, moderate, and high levels of all the three factors of GA, TS, and PP, pairs of them, and a single factor. The three levels of each factor were represented numerically from a uniform scale of 0 (lowest) and 100 (highest) which made the GA, TS, and PP factors seemingly equivalent to start with. Adults judged how favourable those situations were for their respective leaders along a 21-point scale. In Figure 5.3, the solid curves are based on information about two factors, and the dashed curves are based on only one information listed on the horizontal axis. The top graphs are from the main three-cue design and the bottom ones are from the corresponding two-cue designs.

Three results are notable. First, all six sets of solid curves are parallel, supporting the additive model. Second, the separation between solid curves is less in the top graphs than that in the bottom graphs. Because all three top two-way plots are averaged over the levels of the third factor, this lesser separation supports the averaging rule. Finally, the dashed single-factor curve crosses over at least one of the solid curves. Collectively, then, judgements of situational favourableness followed the averaging rule (Singh *et al.,* 1978).

Job Attractiveness and Satisfaction

In the two-factor theory of job satisfaction (Herzberg *et al.,* 1959), job factors are divided into *context* (for example, pay and working conditions) and *content* (for example, nature of task and recognition) categories. Moreover, the context and content factors influence job satisfaction in a non-additive way. Contrary to this prescription, the factorial plots of the Context × Content effects on expected job attractiveness and satisfaction had the pattern of parallelism (Singh 1975).

To choose between the adding and averaging rules, Dalal and Singh (1986) crossed the proportion of good to bad context factors with the proportion of good to bad content factors. Also, they obtained judgements of expected job attractiveness and satisfaction from information presented about either context or content factors. In both experiments, judgements of job attractiveness and expected satisfaction from the two kinds of information had the pattern of parallelism. More important, judgements from any one information crossed over at least one of the curves based on both kinds of information. So, results supported the averaging rule.

Subtractive and Ratio Models

Fair Share of Outcome from a Single Input

When two persons work on a joint project and earn a sum of money (T), how should the 'T' be distributed between them? If the rank orders of inputs and outcomes are simply preserved in outcome allocation (Homans 1961), the rule is *subtractive* or *difference*.

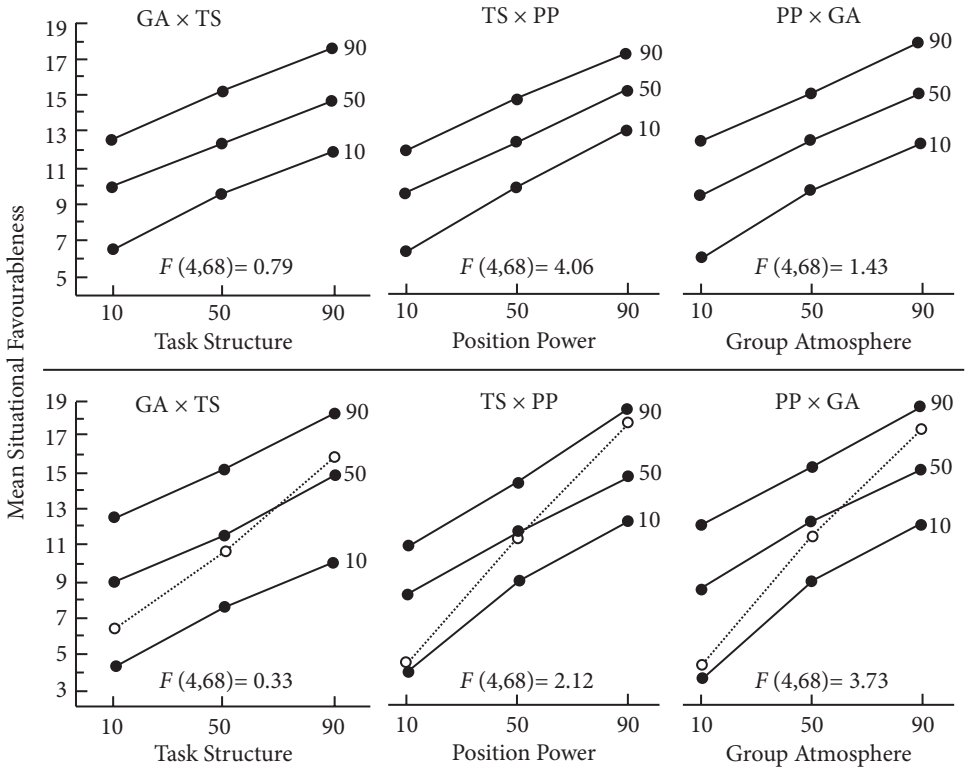

Figure 5.3 Mean Situational Favourableness as a Function of Group Atmosphere (GA), Task Structure (TS), and Position Power (PP) of the Leader

Source: Singh, Bohra *et al.* (1979). Reproduced with Permission from Wiley-Blackwell.
Note: The three two-way plots of the top graphs are from the main three-way Group atmosphere × Task structure × Position power design; and the bottom graphs are from the corresponding two-factor designs of Experiment 3. The dashed curve is based on only the factor listed on the horizontal axis.

However, if each contributor receives the outcome in proportion to what he or she contributed to the project, the rule is *ratio* (Adams 1965).

Singh (1985) formed twenty-five two-worker groups by crossing their respective five levels of performance in a 5 × 5 design. The same allocation task was given to a group of students and managers. Mean allocations by students and by managers are displayed in Figures 5.4 and 5.5, respectively. In both cases, reward for Worker B varies as a positive function of his performance but as an inverse function of the performance of his Co-Worker A. Importantly, the five curves of Figure 5.4 are parallel; those of Figure 5.5 are non-parallel. The vertical difference between the top and bottom curves of Figure 5.5 were 6.06, 7.27, 8.57, 7.58, and 6.10 from left to right, forming the *slanted-barrel pattern* (Anderson 1976).

The pattern of parallelism in Figure 5.4 illustrates subtraction:

$$O_B = T/2 + w\,(I_B - I_A), \tag{2a}$$

where I and O are input and outcome; A and B are the two workers; and w is the importance of the difference between their respective

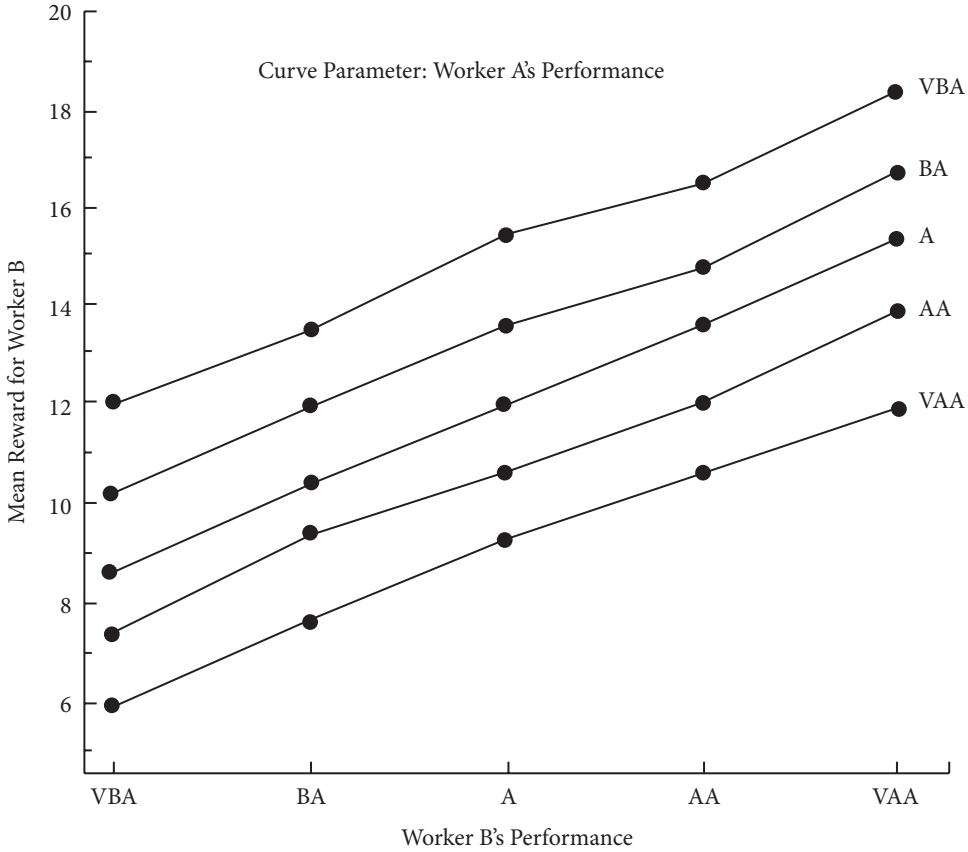

Figure 5.4 Mean Reward to Worker B as a Function of his Co-Worker A's Performance (curve parameter) and His Own Performance by Students (listed on the horizontal axis)

Source: Singh (1985). Reproduced with permission from Taylor & Francis Ltd., www.informaworld.com.
Notes: VAA—Very much above average
AA—Above average
A—Average
BA—Below average
VBA—Very much below average

inputs.[2] In contrast, the slanted-barrel pattern of Figure 5.5 follows from the ratio rule:[3]

$$O_B = [I_B / (I_B + I_A)]\ T. \qquad (2b)$$

Fair Share of Outcome from Multiple Inputs

When students were given information about attitude and performance of Workers A and B, the design became four factorial (Singh 1983). The ratio rule of Equation 2b predicted the slanted-barrel pattern in the A's attitude × B's attitude and A's performance × B's performance effects, as in Figure 5.5, but the pattern of parallelism in the remaining four two-way effects, as in Figure 5.4. Results from four experiments were as expected, illustrating a compound ratio-averaging model of outcome allocation.

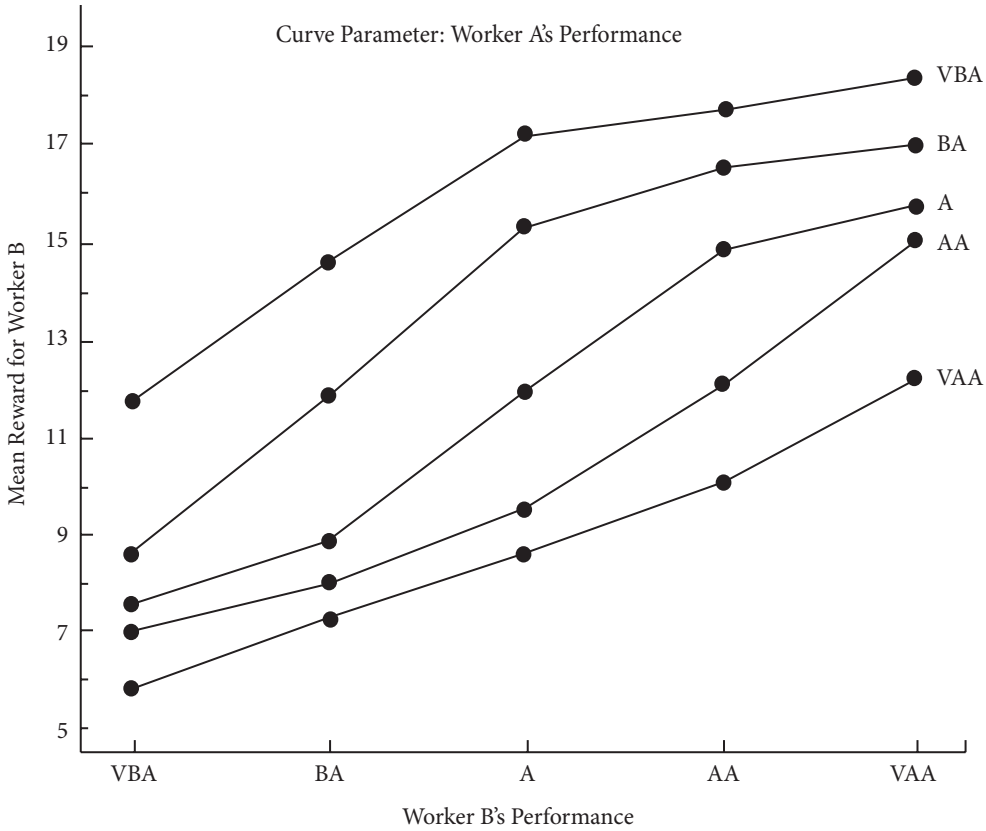

Figure 5.5 Mean Reward to Worker B as a Function of his Co-Worker A's Performance (curve parameter) and His Own Performance by Managers (listed on the horizontal axis)

Source: Singh (1985: Experiment 5). Reproduced with permission from Taylor & Francis Ltd., www.informaworld.com.

Disciplinary Actions in Organizations

Adults learned about seriousness of rule infractions in an organization and about the job performance of the accused employees and initiated disciplinary actions against them. One of the responses sought was punishment which varied as a positive function of seriousness of transgression but as an inverse function of level of job performance (Singh 1978). The pattern of parallelism supported the subtractive rule:

Punishment = Seriousness of transgression – Level of performance. (3)

Adding versus Averaging with Imputations about Missing Information

Predictions of Performance

Heider (1958) proposed that people use the multiplicative rule in predicting performance from information about motivation and ability. If he is right, predicted performance should have a linear fan pattern as in the rightmost graph of Figure 5.2.

In Experiment 2 of Singh, Gupta *et al.* (1979), however, the four solid curves from the motivation and ability information and

the dashed curve from the ability information alone, shown in the top graphs of Figure 5.6, were essentially parallel. At the overt level, this result suggested the adding rule:

Performance = Motivation + Ability. (4)

The averaging results reported in Table 5.1 and displayed in Figure 5.2 assumed that judgements depend upon only the information given. The crossovers of the two-cue curves by the single-cue curves in Figure 5.3 attest the soundness of this assumption. If this assumption is taken strictly, then the results in the top graphs of Figure 5.6 have to be accepted as the evidence of the adding rule of Equation 4. With an alternative assumption that participants imputed a constant value to the missing motivation information and averaged this imputation with the given ability information (Singh 1991), however, the averaging rule can also account for the parallelism obtained.

Weight as Another Component of Valuation

Findings presented heretofore indicate that the given information is always converted into psychological quantity. Each such psychological quantity has a scale value (*s*) and a weight value (*w*). In an integration model, the *s* places the information along the judgement measure, and the *w* represents the relative importance of each piece of information in the overall judgement.

Consider, for example, the experiments on judgements of favourableness of leadership situations (Singh, Bohra *et al.*, [1979]). The GA, TS, and PP factors of situations in Figure 5.3 had uniform numerical values of 10, 50, and 90 that supposedly represented them uniformly along the response measure of situational favourableness. Nevertheless, the same GA was most effective when it was presented alone (see the dashed curve), less effective when paired with TS (see the bottom set of three solid curves), and least effective when paired with both TS and PP (see the top set of three solid curves) in the left panel. Similar trends hold with the effectiveness of the TS and PP factors of Figure 5.3. These differences in the effect of the very same information across the three designs indicate that the greater the amount of information, the lesser the *w* of any one piece of information. Accordingly, both the *s* and *w* parameters are necessary and important in psychological quantification of any information (Anderson 1981).

Singh and Bhargava (1986) investigated: (a) the interplay of *s* and *w* within the averaging model for predictions of performance by manipulating reliability of motivation and ability information; and (b) imputations, if any, about missing information by taking judgements from information presented about either motivation or ability. Reliability of information affects its weight (Anderson 1971; Birnbaum *et al.*, 1976). Thus, the greater the reliability of motivation and ability information, the greater should be their respective effects on predicted performance.

It is the relationship between the reliability of information of one kind and the value of information of another kind that is critical for diagnosing the exact form of averaging.[4] If the relative-weight averaging model applies, then the greater the reliability of information of one kind, the less should be the effect of the value of information of another kind (Surber 1981a). When the constant-weight averaging model holds, however, the effect of value of information of one kind should be independent of reliability of information of another kind (Birnbaum 1976; Birnbaum *et al.*, 1976).

The relationships between the reliability and value of information in predicted

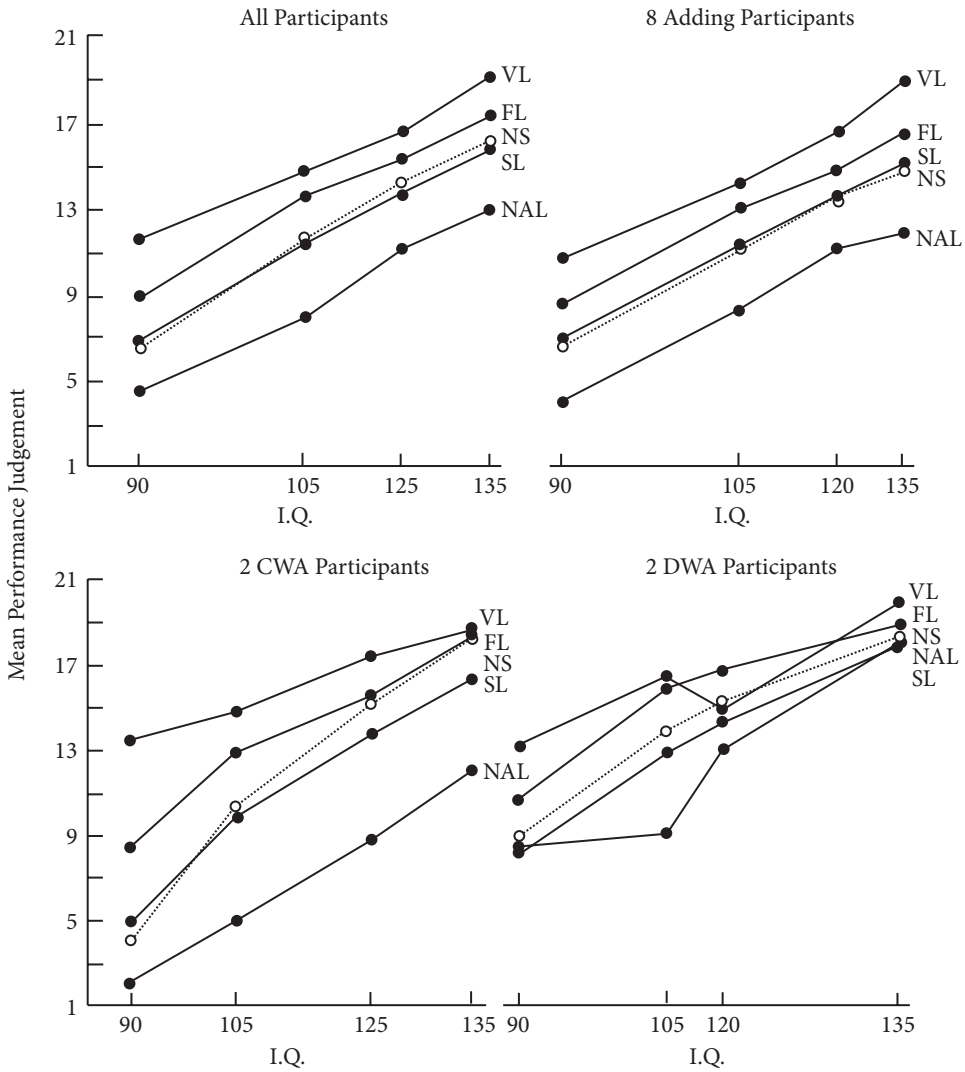

Figure 5.6 Mean Predicted Performance as a Function of Labouriousness (curve parameter) and IQ (listed on the horizontal axis)

Source: Singh, Gupta *et al.* (1979). Reproduced with permission from American Psychological Association (APA).

Notes: (a) The levels of labouriousness were: NAL, not all labourious; SL, slightly labourious; FL, fairly labourious; VL, very labourious; NS, labouriousness not specified.

(b) The classification of the participants into followers of the adding, constant-weight averaging (CWA), and differential-weight averaging (DWA) rules was based on results from single-participant analyses.

(c) Results in the bottom two graphs are not of immediate interest.

performance are displayed in Figure 5.7. The first four graphs from left—two from the single-cue motivation (Motivation reliability × Motivation) or ability (IQ reliability × IQ) designs and two from the 2-cue design (Motivation reliability × Motivation × IQ

reliability × IQ design)—show that the greater the reliability of information of one kind, the greater its effect on predicted performance. The crossovers of the curve representing the information of low reliability by the curve representing the information of high reliability in each of the four graphs also agree with the general averaging theory. However, the patterns of parallelism in the two rightmost graphs, namely, the Reliability of ability information × Value of motivation information effect and the Reliability of motivation information × Value of ability information effect, illustrate the independence of the effect of value of information of one kind from the effect of reliability of information of another kind. Hence, results supported the constant-weight averaging model in predictions of performance (Singh, Gupta *et al.*, 1979).

Figure 5.8 plots the two three-way interaction effects, namely, the Motivation reliability × Ability × Motivation effect in the two left graphs and the Ability reliability × Motivation × Ability effect in the two right graphs. The dashed curve is based on motivation or ability information alone listed on the horizontal axis. The patterns of parallelism in the four solid curves of the four graphs reconfirm the constant-weight averaging model. The dashed motivation-only curve of the two left graphs has steeper slope than any of the solid curves, implying the averaging process as in Figure 5.3. However, the dashed ability-only curve of the two right graphs is closer to the

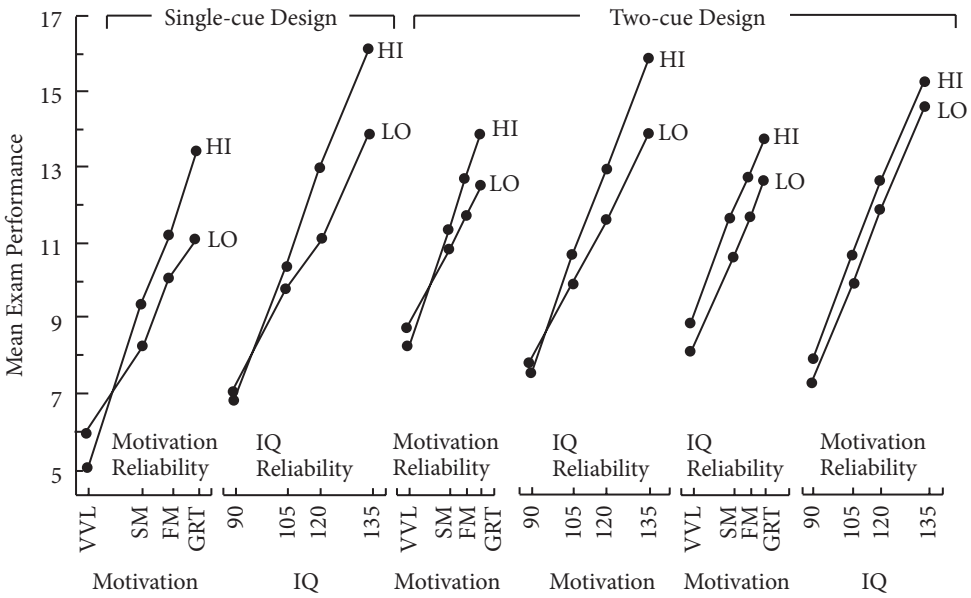

Figure 5.7 Two-way Factorial Plots of the Motivation Reliability × Motivation Effect (the first and third graphs), Ability Reliability × Ability Effect (the second and fourth graphs), Ability Reliability × Motivation Effect (the fifth graph), and Motivation Reliability × Ability Effect (the rightmost graph)

Source: Singh and Bhargava (1986). Reproduced with permission from Elsevier.
Note: The first two graphs are from the single-cue Motivation reliability × Motivation and Ability reliability × Ability designs; the remaining four are from the two-cue design.

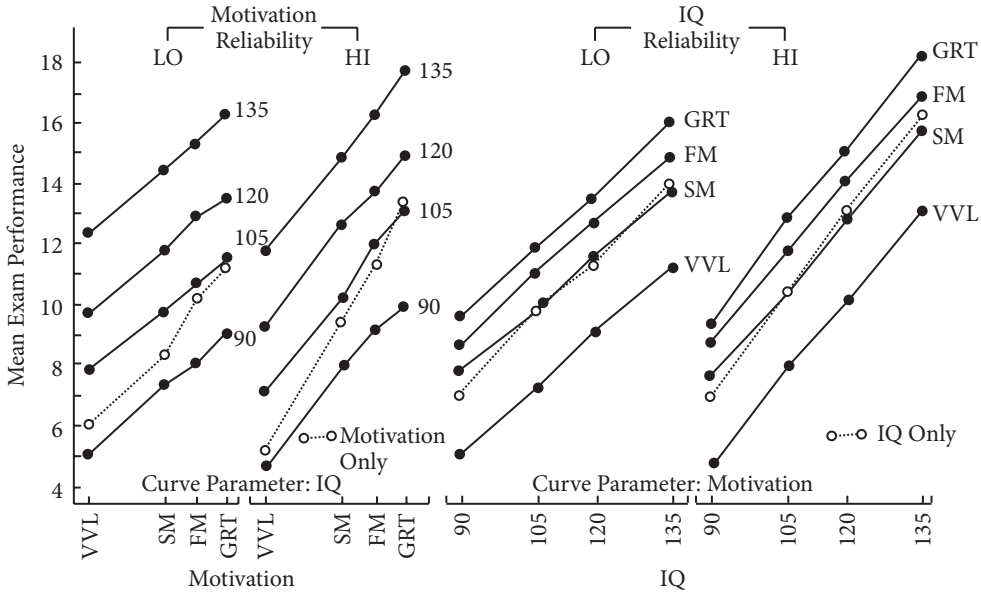

Figure 5.8 Two-way Factorial Plots of the Ability × Motivation Effect under Low and High Conditions of Motivation Reliability and of the Motivation and Ability Effect under Low and High Conditions of Ability Reliability

Source: Singh and Bhargava (1986: Experiment 2). Reproduced with permission from Elsevier.
Notes: (a) The levels of motivation were: VVL, very very low; SM, some; FM, fairly much; and GRT, a great deal.
(b) The dashed curve is from either motivation or ability information presented alone.
(c) The two left graphs represent the Motivation reliability × Ability × Motivation effect; and the two right graphs represent the Ability reliability × Motivation × Ability effect of the main four-cue design.

solid curve of slightly motivated, implying the adding process of Equation 4 as in the top two graphs of Figure 5.6.

Results of Figure 5.8 are seemingly inconsistent. The left two graphs suggest an averaging rule, but the right two graphs argue for an adding rule. If the possibility of imputations about missing information is accepted, then the interpretations of the effects of the information about either motivation or ability presented alone become simpler. Given the evidence for the constant-weight averaging rule, it is likely that the missing ability information was imputed a value that directly varied with the value of given motivation information but the missing motivation was imputed a constant value. Had such imputations not occurred, the dashed curve of the two right graphs should have been at the bottom as in the leftmost graph for the adding rule of Figure 5.2. By allowing imputations, therefore, the results of all the four graphs of Figure 5.8 are interpretable as cases of the same averaging rule but with different imputations about missing ability or motivation information.

Multiplying Model

Predictions of Gift Size

Generosity, like motivation, is an energizing factor; income, like ability, is a capability

factor (Graesser and Anderson 1974). Will information about generosity and income be then integrated by the averaging rule or the multiplying rule? To address this question, Singh (1991) manipulated the reliability and value of information about both generosity and income of targets, and sought predictions of the size of gift they might send to a needy family. The predictions from the two forms of averaging model (see Endnote 4) were precisely like those for predictions of performance discussed earlier. However, the multiplying model took the form of:

$$\text{Gift size} = w_G G \times w_I I, \quad (5)$$

where G and I are values of the generosity and income factors and w_G and w_I are their respective weights. In this case, an increase in the reliability of generosity information should amplify the effect of not only the value of generosity information but also of the value of income information and vice versa.

Figure 5.9 presents the Generosity × Income effects from the various combinations of the reliability of information about generosity (listed from left to right) and income (listed from bottom to top). Three results are striking. First, every graph has the linear fan pattern. Second, the difference between curves, that represents the effect of generosity, increases as the reliability of generosity information increases from left to right. Similarly, the slope of the income curves increases as the reliability of income information increases from top to bottom. Finally, and the most important, the effects of values of the two kinds of information are stronger when they came from the sources of high reliability (see the bottom right graph) than when they came from the sources of low reliability (see the upper left graph). Thus, results illustrated the cognitive algebra of the equation of:

$$\text{Gift size} = \text{Generosity} \times \text{Income}.$$

Recall that motivation and ability information were averaged together in predictions of performance. Averaging implies that the two factors were construed as multiple sufficient causes (Surber 1981b). When imputations about missing motivation or ability information can be as pervasive in predictions of performance as illustrated earlier, similar imputations can be stronger in predictions of gift size because the multiplying rule stands for the multiple necessary causal schema (ibid.).

Imputations about Missing Income Information

In three experiments, Singh (1991) crossed two pieces of information about generosity with a single piece of information about income. Predictions of gift size were also sought from one of those pieces of generosity information. The hypothesized cognitive algebra was:

$$\text{Gift size} = [(\text{Generosity}_1 + \text{Generosity}_2)/2] \times \text{Income}. \quad (6)$$

Results showed that the effects of two pieces of generosity information on the predicted gift size had the pattern of parallelism. However, the Income × Generosity_1 and Income × Generosity_2 effects had the linear fan pattern. The curve based on either generosity information crossed at least one curve based on both generosity information which means that they were averaged together. Results illustrated the compound averaging–multiplying model of Equation 6.

Figure 5.10 presents the mean predicted gift size as a function of income and generosity information. The generosity levels of the 3 × 3 design on the horizontal axis are spaced according to their functional values to make the linear fan pattern stand out. Evidently, the solid curves from all three experiments

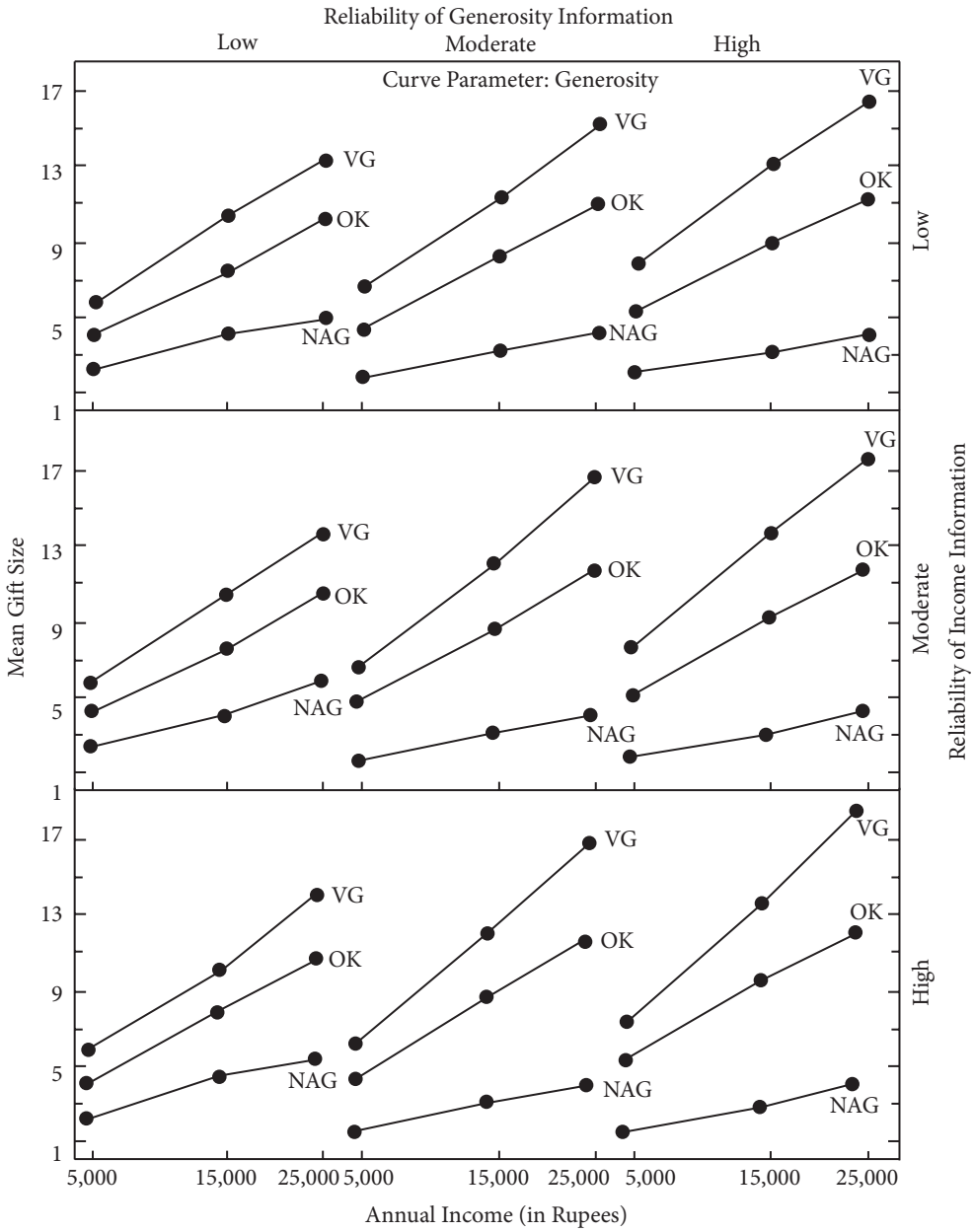

Figure 5.9 Factorial Plots of the Generosity × Income Effects under the Nine Conditions of Reliability of Generosity and Income Information

Source: Singh (1991: Experiment 6). Reproduced with permission from Taylor & Francis Ltd., www.informaworld.com.

Notes: (a) NAG, Not at all generous; VG, Very generous.

(b) Reliability of information about generosity increases from the left to the right side; that of information about income increases from the top to the bottom.

have the linear fan pattern, and the dashed generosity-only curve crosses over at least one solid curve. Given the evidence for the multiplicative model, this crossover of the solid curve by the dashed one reflects imputations about missing income information. Specifically, the missing income information, like missing ability information in prediction of performance (Singh and Bhargava 1986), was imputed a value that increased with the value of generosity (or motivation) information given.

Imputations about Missing Generosity Information

Figure 5.11 displays the joint plots of the two-way Generosity$_1$ × Income, Generosity$_2$ × Income, and Generosity$_3$ × Income effects from an experiment that had the four-cue Generosity$_1$ × Generosity$_2$ × Generosity$_3$ × Income factorial design and the corresponding two-cue factorial designs. Predictions from information about income alone are shown by the dashed curve.

Results first confirm the previous findings of Gift size = Generosity × Income. The linear fan pattern is present in the separate as well as joint plots of the results from the four- and two-cue factorial designs. The slopes of the curves from the four-cue design that are averaged over the other two generosity cues lie between the slopes for the low and high generosity curves from the two-cue designs. Such positions of the solid curves show that information integration was again by a compound averaging–multiplying model as in Equation 6. Because the dashed curve of each graph fits well within the linear fan pattern

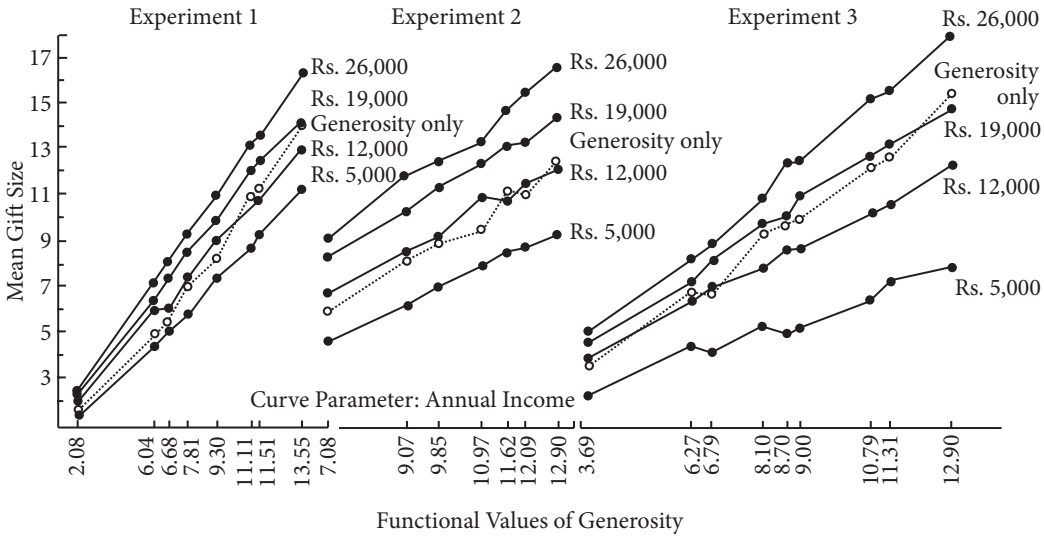

Figure 5.10 Mean Judgement of Gift Size as a Function of Income and Generosity Information

Source: Singh (1991). Reproduced with permission from Taylor & Francis Ltd., www.informaworld.com.
Notes: (a) Data from Income × Generosity$_1$ × Generosity$_2$ design of Experiments 1–3.
(b) The nine levels of the 3 × 3 Generosity$_1$ × Generosity$_2$ component of the three-factor design are spaced on the horizontal axis according to their functional values.
(c) The dashed curves are from the two-cue Generosity$_1$ × Generosity$_2$ designs of the three respective experiments.

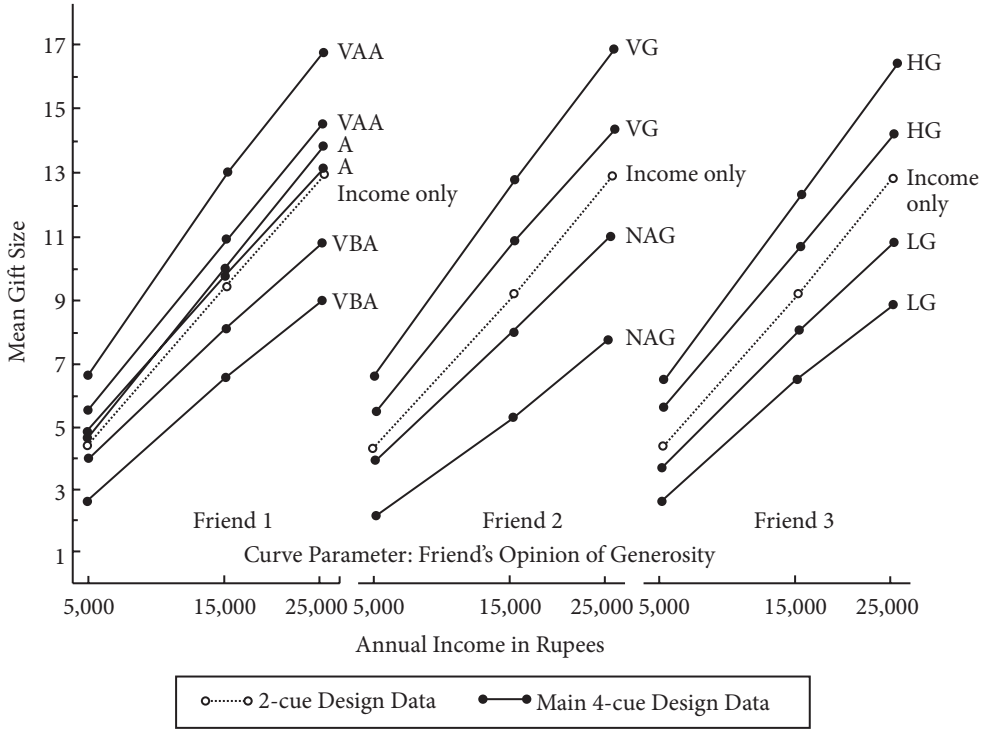

Figure 5.11 The Generosity$_1$ × Income, Generosity$_2$ × Income, and Generosity$_3$ × Income Effects from the Four-cue design (solid curves with open circles) and from the Corresponding Two-cue Designs (solid curves with filled circles)

Source: Singh (1991: Experiment 4). Reproduced with permission from Taylor & Francis Ltd., www.informaworld.com.
Notes: (a) NAG, Not at all generous; VG, Very generous; LG, Low in generosity; HG, High in generosity.
(b) Dashed curve represents judgements based on information about income only.

and that of the leftmost graph is no different from the curve for the given average generosity information, the missing generosity information was imputed a constant value of average (Singh 1991).

Considered together, findings indicate that imputations about missing information are more likely in predictions than in evaluations. In predictions, moreover, imputations are made regardless of the cognitive algebra of averaging, as in predictions of performance (Singh and Bhargava 1986), or of multiplying, as in predictions of gift size (Singh 1991).

Summary

The evidence presented so far demonstrates that information integration is indeed a basic cognitive process in Indians. Contingent upon the demands of judgements asked and the information presented, Indians render judgements that obey the averaging, subtracting, multiplying, and dividing rules. When tasks present complex sets of information, such as in reward allocation from multiple inputs (Singh 1983) and predictions of performance (Singh and Bhargava 1986) or gift size (Singh 1991) from multiple pieces of information

about motivation and generosity, information integration follows the correspondingly complex models of averaging–ratio rules or averaging–multiplying rules.

In predictions, information integration is not confined to only what is presented. The missing motivation or generosity information is imputed a constant value, but the missing ability or income information is imputed a value that varies directly with the value of the given motivation or generosity information. Apparently, Indians have different representations of the relationships between the missing and given information about energizing (for example, motivation and generosity) and capability (for example, intelligence and income) forces within an individual person.

Comparisons with American Results

To what extent are the foregoing information integration findings unique to Indians? The literature has several integration–theoretical experiments conducted exclusively in India or the US. Because those experiments are conceptually comparable across cultures (Ember 1977; Malpass 1977), it is possible to address the issues of generality and specificity of information integration by Indians.

Generality in Moral Domains

Farkas and Anderson (1979), who studied equity judgements from information about effort and performance of two persons, or from one of those two inputs, reported evidence for the difference, ratio, and compound averaging–ratio models as in Singh (1983, 1985). Equity judgements by even American kindergartners followed the ratio rule (Anderson and Butzin 1978).[5] Results showed that the belief about what is a *fair* share in an exchange fully develops by 4 years of age and remains so thereafter.

The research programme on distinguishing multiplying versus averaging in predictions of gift size from generosity and income information (Singh 1991) was prompted by the American finding that Gift size = Generosity × Income (Graesser and Anderson 1974). Eight experiments with different kinds of manipulations in India not only obtained the same linear fan pattern in the Generosity × Income effect as did the two experiments in the US but also demonstrated that the integration rule was multiplying (see Figures 5.9 and 5.11).

Although the experiments on judgements of equity and gift size were not methodologically comparable, results from both Indian and American participants were similar. Such convergent results point out that information integration in moral judgements is as basic in India as in the US. Theoretically, therefore, the cross-age and cross-cultural similarities in moral beliefs uncovered by integration–theoretical analyses had hinted at the same moralistic streak in human nature as proposed later by evolutionary psychologists (Fehr and Fischbacher 2004).

Specificity in Achievement Domains

In predictions of performance from motivation and ability, information integration by Indians appears to be unique in three notable ways. First, predictions of academic performance by Indian adults followed the constant-weight averaging model (Singh and Bhargava 1986; Singh, Gupta *et al.*, 1979) but those by American adults followed the relative-weight averaging model (Surber 1981a). So, cultural differences are in the form of the averaging model.

Second, difficulty of exam moderated the pattern in the Motivation × Ability effect in the US (Surber 1981b). Specifically, the interaction effect had the linear fan, parallelism,

and inverted linear fan (convergence towards the right) patterns when exams were extremely difficult, OK, and easy, respectively. However, there was no moderation of the pattern of parallelism by exam difficulty in India (Singh and Bhargava 1985). In prediction of life performance, opportunity for growth—a factor conceptually related to task difficulty—did not moderate the pattern in the Motivation × Ability effect either (Singh 1988). Predicted exam performance was higher when exam was easy than difficult, and life performance was higher when opportunity for growth was great than very little. Thus, the cultural difference is in integration of information about task difficulty.

Finally, predictions by Americans adults yielded the linear fan pattern in the Motivation × Ability effect regardless of the achievement areas (Anderson and Butzin 1974; Kun *et al.*, 1974; Surber 1980, 1981b). However, such pattern was rather absent in Indian studies of adults. Hence, there are cultural differences in the pattern in the Motivation × Ability effect.

Cultural Difference Hypothesis

It was the pattern difference in the Motivation × Ability effect on predicted performance by the American and Indian adults that led Singh, Gupta *et al.* (1979) to propose the cultural difference hypothesis. Given the achievement orientation of American culture, the authors argued, the linear fan pattern of the multiplying rule may imply an 'elitist' causal belief of American adults that effort or trying can be more effective with persons of higher ability than those of lower ability. In contrast, the pattern of parallelism in predictions of performance by Indian adults may imply an 'egalitarian' causal belief that '… each person, regardless of native ability, has equal opportunity to improve his or her lot' (ibid.: 1350).

Kun *et al.* (1974) explained the development of the linear fan pattern from the pattern of parallelism in the US by the *cognitive capacity* to use the complex multiplicative rule later than the simple additive rule. An unfortunate implication of this account was that Indians adults had to be regarded as *cognitively deficient*, a conclusion that would be clearly at odds with the evidence for the use of the multiplying or dividing rule by Indian participants in other tasks (Singh 1983, 1985, 1991).

Singh, Gupta *et al.* (1979) proposed, therefore, that the locus of India–America difference in prediction of performance is in culture-specific causal beliefs about how motivation and ability determine performance. An advantage with this view was that it interpreted the patterns of parallelism, linear fan, and inverted linear as the *variety* (Ember 1977), instead of *complexity* (Kun *et al.*, 1974), in cognitive process across cultures.

Culture as a Moderator

Results of Surber (1981a, 1981b, 1985) raised doubt against the cultural difference in the use of averaging rule but not against the interpretation of cognitive algebra as causal beliefs. In fact, the patterns in predicted exam performance by American adults agreed with their self-reported causal beliefs. In Surber (1981b), for example, participants believed that either motivation or ability was *sufficient* for doing well at an easy task but both motivation and ability were *necessary* for doing well at an extremely difficult task. Consistent with these beliefs, the Motivation × Ability effects on predicted performance in difficult and easy exams had the linear fan and inverted linear fan patterns, respectively. An inverted linear fan pattern implies the Horatio Alger belief that by trying hard anyone can succeed (Singh and Singh 1994; Srivastava and Singh 1988).

Surber (1985) classified American participants into believers of a greater importance of motivation (motivation-importance) or of a greater importance of ability (ability-importance) in leading to achievement. The effect of the motivation information on predicted performance was stronger in motivation-importance believers than in ability-importance believers. By contrast, the effect of the ability information on predicted performance was stronger in ability-importance believers than in motivation-importance believers. More interestingly, the differences in the effects of the two kinds of information among the two groups of believers were via changes in weights, not values, of the averaging model. Thus, people of different cultures may use the same averaging rule but weight motivation and ability information uniquely as per their belief systems (Hau and Salili 1996).

Given the foregoing results, predictions by Americans and Indians cannot be as clearly linked to the elitist and egalitarian causal beliefs as Singh, Gupta *et al.* (1979) envisaged. Nevertheless, cognitive algebra of predicting performance in India is still more consistent with the egalitarian causal belief than with other causal beliefs. To support this hypothesis, Srivastava and Singh (1988) studied predictions of performance in a singing contest. In singing, ability is the critical factor (Riemer 1975). So, prediction of singing performance by even young Indians may be consistent with the linear fan pattern.

Figure 5.12 presents mean predicted performance in a singing contest from motivation and ability information by participants

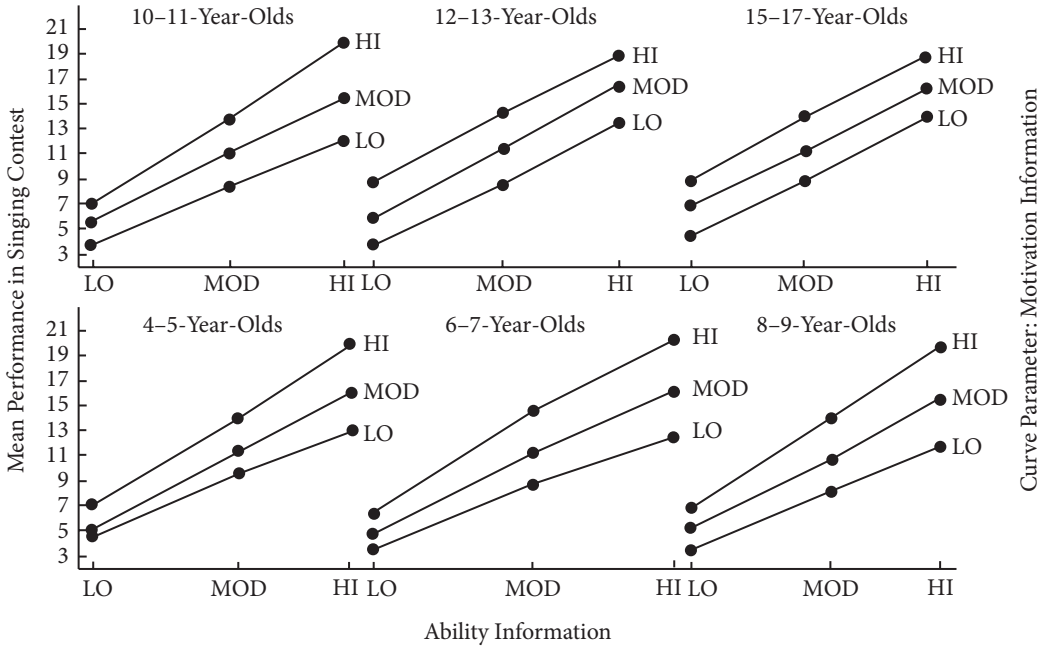

Figure 5.12 Mean Performance in a Singing Contest as a Function of Motivation and Ability Information and Age of the Participants

Source: Srivastava and Singh (1988: Experiment 2). Reproduced with permission from Wiley-Blackwell.
Notes: LO, Low; MOD, Moderate; HI, High.

of six age groups. The three bottom and top leftmost graphs are for 4–5, 6–7, 8–9, and 10–11-year-olds. All of them have a uniform linear fan pattern as if the multiplicative rule were operative. In contrast, the centre graph for the 12–13-year-olds and the right graph for 15–17-year-olds have the pattern of parallelism. This age-related shift from the linear fan pattern to the pattern of parallelism indicates that the additive rule replaces the multiplicative rule by the beginning of adolescence in India, a result opposite of that from the US (Kun *et al.*, 1974; Surber 1980). Individual child analyses disclosed several transitions from the additive to the multiplicative model and vice versa but the additive model had replaced the multiplicative one firmly by teenage. Evidently, the capacity to make predictions according the elitist belief implied by the multiplicative rule develops earlier in India than in the US but the egalitarian belief rebounds by the onset of adolescence.

Singing is secondary to academic achievement. Even before doing well in singing, one has to complete his or her studies. Would predictions of exam performance that shapes one's career conform to the pattern of parallelism at even younger ages? Gupta and Singh (1981) reported evidence for development of the pattern of parallelism from the inverted linear fan pattern (that is, the Horatio Alger belief) by age of 8 years. To further confirm the cultural emphasis on an egalitarian causal belief in academic achievement, Singh and Singh (1994) asked children between the ages of 4 and 10 years to predict exam performance of students, using information about both motivation and ability. Each kind of information came from two independent sources: motivation information from mother and neighbour; and ability information from two different teachers of the same school. Stimulus descriptions had four cues as well as two cues. Unlike in the previous Indian studies, children made repeated predictions over five trials spaced over three days that allowed an assessment of whether cognitive capacity to integrate information is already present at age four (Anderson and Butzin 1978) or is induced by trials of judgements (Gelman 1978).

Figure 5.13 displays the four two-way plots of the Motivation × Ability effects (that is, Mother × Teacher 1, Neighbour × Teacher 1, Mother × Teacher 2, Neighbour × Teacher 2 from left to right) from the four-cue design. Predictions by participants from five age groups are shown from bottom to top. Notably, predictions of exam performance conformed more and more to the pattern of parallelism as children aged.

Individual child analyses further revealed that even kindergarteners used all the four given pieces of information from the very beginning. The cue usage in predicted performance was better when there were four cues than when there were only two cues. This demonstration suggests that poorer performance at younger ages in American studies (Anderson and Butzin 1978; Kun *et al.*, 1974) might have been because of the use of two-cue tasks and/or the insufficient data from each individual child.

The foregoing developmental findings indicate that Indian adolescents predict performance according to the egalitarian belief regardless of the nature of task. At younger ages, the nature of task does engender different patterns in the Motivation × Ability effect (Srivastava and Singh 1988), implying the presence of the Horatio Alger, elitist, and egalitarian beliefs among them. However, those patterns shift to a uniform pattern of parallelism with age. Such shift occurs earlier in prediction of academic achievement than of non-academic achievements. Thus,

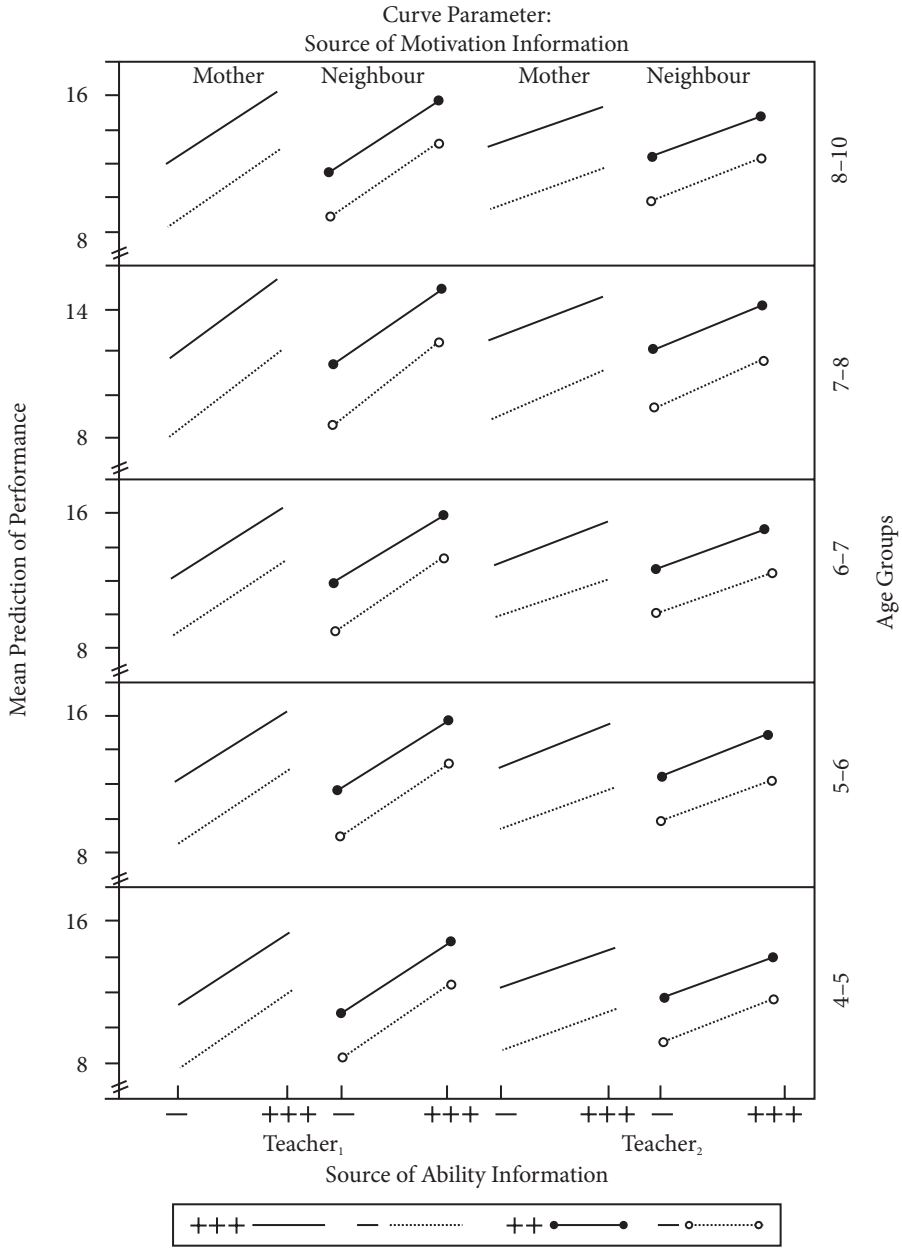

Figure 5.13 Factorial Plots of the Mother × Teacher$_1$, Neighbour × Teacher$_1$, Mother × Teacher$_2$, and Neighbour × Teacher$_2$ Effects on Predicted Exam Performance by the Children of Five Age Groups

Source: Singh and Singh (1994). Reproduced with permission from Elsevier.

Note: The sources of information about motivation (that is, how much time the child spent studying at home) were the mother and the neighbour; those of information about ability (that is, potential for learning or brightness) were two school teachers.

culture remains a moderator of information integration in prediction of performance, using motivation and ability, and the information integration in achievement judgements by Indians is indeed unique.

General Considerations

This chapter has achieved its central task of documenting that information integration is a basic cognitive process. When Indians are provided with multiple pieces of information about individual persons, social groups, job offers, and leadership situations, for example, and asked to make judgements, their ways of information integration conform to cognitive algebra of adding, averaging, dividing, multiplying, subtracting, or a combination of them. When the needed information is missing as in predictions of performance and gift size, they integrate information is ways that suggest that they had first imputed values to missing information from the given information. Of those rules, averaging is the most frequently used one. Nevertheless, activation of an integration rule is driven by the nature and amount of information and the judgements solicited. Clearly, *purpose* governs an ongoing thought.

Given the evidence for judgements by children, adolescents, and managers of India obeying the prescriptions of cognitive algebra, three observations are in order. First, children as young as 4 years of age (Singh and Singh 1994; Srivastava and Singh 1988) are adept in converting physical stimuli into psychological quantities. Moreover, the psychological quantities they use in rendering the judgements can be estimated through the integration rule as in functional measurement (Anderson 1982). This power of the functional measurement was specifically illustrated by Singh, Bohra *et al.* (1979) in the case of ordering of leadership situations. Plots of the leadership effectiveness indices reported by Fiedler (1971) against the rank orders and functional values of situational favourableness disclosed that the support for the contingency model was much better with values from functional measurement than with simply the rank orders (see Figure 5 in Singh, Bohra *et al.*, 1979: 261). Apparently, functional measurement encourages taking up old problems in the literature and coming up with solutions much better than those already accepted.

Second, cognitive algebra represents causal beliefs, not mathematical calculations. Judgements of playgroup and job attractiveness followed the averaging rule because the given pieces of information are believed to be compensatory; equity judgements obeyed the ratio rule because people believe that outcome should be proportional to input; and judgements of gift size followed the multiplying rule because people believe that generosity can be more effective with rich rather than poor in contributing to charities. The most convincing evidence for the belief interpretation of cognitive algebra was the difference between the integration rule for predictions of performance from motivation and ability information and predictions of gift size from generosity and income information. In spite of the conceptual similarity between the energizing forces of motivation and generosity and capability forces of ability and income (Graesser and Anderson 1974), cognitive algebras of the equations of Performance = Motivation × Ability and Gift size = Generosity × Income were averaging and multiplying, respectively. Imputations made about missing motivation and generosity and about missing ability and income also reflected their beliefs about how the given and missing information were causally linked. Whereas imputations about missing ability

and income were positively correlated with given motivation and generosity information, imputations about missing motivation and generosity from given ability and income information were independent.

Finally, the integration rule is an ideal criterion for any comparative study. As Gupta and Singh noted:

> An important advantage with the integration rules is that they deal with the *pattern* of responses and not with the numerical value of single responses. This aspect is vital for comparison between different ages or cultures. No *a priori* knowledge of value of stimuli or origin and unit of personal response scale is required. Groups as well as individuals are readily comparable with respect to the pattern in their judgments of stimuli constructed from factorial design. The integration rule also permits comparison of groups along the criteria of information utilization and information valuation. (1981: 824)

The second task of this chapter was to highlight the foregoing potential of the integration rule. Given the empirical results summarized in this chapter, the second mission also seems to have been attained in three ways. First, although the tasks for predicting performance from motivation and ability information were not methodologically comparable across the Indian and American studies, the criterion of integration rule did uncover important cultural differences in belief systems. Second, the causal beliefs of Indian children are as diverse as those of American adults. A greater diversity in casual beliefs of children than of adults points to a pressure on endorsing a uniform causal belief in Indian culture. The opposite pattern in causal beliefs of Americans, however, suggests that American culture promotes a greater diversity among adults than children. It would be premature, therefore, to regard any of the patterns in the Motivation × Ability effect as a *mature* way of information integration (Srivastava and Singh 1988). Finally, it was the integration rule that allowed diagnosis of the patterns of imputations about missing information. That missing ability and income information were imputed values that increased with the value of given motivation and generosity information suggests that capability is malleable (Dweck and Elliott 1983) to Indians. Perhaps they believe that one can change one's capability, either intelligence or income, through putting in effort or being generous.

Notes

1. In information integration experiments, there are usually more stimuli than those formed by the Row × Column design. Filler stimuli use levels other than those in the main stimuli to disguise the transparency of the hypothesis, and extreme stimuli serve as anchors for defining the end points of the response measure. Participants are also given practice with a sample of the stimuli to enable them develop a clear idea of the task and the response measure.

2. The weight represents the importance of the difference between the inputs of the two persons. Even when persons A and B differ considerably in their contributions, the difference might be ignored. Under such circumstance, both persons would receive an equal outcome (T/2). So, the seemingly parity rule is a result of the same difference rule but with a zero weight to the difference between inputs of the two workers.

3. This ratio rule is a special case of averaging theory (Anderson 1981).

4. One form of the averaging model is $J = (w_O s_O + w_A s_A + w_B s_B) / (w_O + w_A + w_B)$, where J is judgement, s_O, s_A, and s_B are the respective scales values of a subjective factor of initial opinion of the participant and of external row Factor A and column Factor B, and w_O, w_A, and w_B are their absolute weights, respectively. The relative weights of Factor A and Factor B are $w_A/(w_O + w_A + w_B)$ and $w_B/(w_O + w_A + w_B)$, respectively. The relative weights are required to sum to 1, the condition for an averaging model (Anderson 1981). In the constant-weight averaging model, $J = [w'_O (w_O s_O) + w'_A (w_A s_A) + w'_B (w_B s_B)]/(w'_O + w'_A + w'_B)$, where w'_O, w'_A, and w'_B are

weights for the three weight-scale value combinations independent of and not affected by information reliability (Birnbaum 1976; Birnbaum *et al.*, 1976).

5. Sinha (1985) reported the pattern of slanted barrel in reward divisions by 5 year old Indian and Nepalese children.

References

Adams, J. Stacy. (1965), 'Inequity in Social Exchange', in Leonard Berkowitz (ed.), *Advances in Experimental Social Psychology*, New York: Academic Press, pp. 267–99.

Anderson, N.H. (1965), 'Averaging versus Adding as a Stimulus-combination Rule in Impression Formation', *Journal of Experimental Psychology*, 70(4): 394–400.

_____. (1971), 'Integration Theory and Attitude Change', *Psychological Review*, 78(3): 71–206.

_____. (1974), 'Cognitive Algebra', in Leonard Berkowitz (ed.), *Advances in Experimental Social Psychology*, New York: Academic Press, pp. 1–101.

_____. (1976), 'Equity Judgments as Information Integration', *Journal of Personality and Social Psychology*, 33(3): 291–9.

_____. (1981), *Foundations of Information Integration Theory*, New York: Academic Press.

_____. (1982), *Methods of Information Integration Theory*, New York: Academic Press.

Anderson, N.H. and C.A. Butzin. (1974), 'Performance = Motivation × Ability: An Integration Theoretical Analysis', *Journal of Personality and Social Psychology*, 30(5): 598–604.

_____. (1978), 'Integration Theory Applied to Children's Judgments of Equity', *Developmental Psychology*, 14(6): 593–606.

Birnbaum, M.H. (1976), 'Intuitive Numerical Prediction', *American Journal of Psychology*, 89(3): 417–29.

Birnbaum, M.H., R. Wong, and L.K. Wong. (1976), 'Combining Information from Sources That Vary in Credibility', *Memory and Cognition*, 4(3): 330–6.

Dalal, A.K. and R. Singh. (1986), 'An Integration–Theoretical Analysis of Expected Job Attractiveness and Satisfaction', *International Journal of Psychology*, 21(4): 555–64.

Dweck, C.S. and E.S. Elliott. (1983), 'Achievement Motivation', in E. Mavis Hetherington (ed.), *Carmichael's Manual of Child Psychology: Social and Personality Development*, New York: Wiley, pp. 643–91.

Ebenbach, D.H. and C.F. Moore. (2000), 'Incomplete Information, Inferences, and Individual Differences: The Case of Environmental Judgments', *Organizational Behavior and Human Decision Processes,* 81(1), pp. 1–27.

Ember, C.R. (1977), 'Cross-Cultural Cognitive Studies', in Bernard J. Siegel (ed.), *Annual Review of Anthropology*, Palo Alto, CA: Annual Reviews, pp. 33–56.

Farkas, A.J. and N.H. Anderson. (1979), 'Multidimensional Inputs in Equity Theory', *Journal of Personality and Social Psychology,* 37(4): 579–96.

Fehr, E. and U. Fischbacher. (2004), 'Social Norms and Human Cooperation', *Trends in Cognitive Science*, 8(4): 87–190.

Fiedler, Fred E. (1967), *A Theory of Leadership Effectiveness*, New York: McGraw-Hill.

_____. (1971), 'Validation and Extension of the Contingency Model of Leadership Effectiveness: A Review of Empirical Findings', *Psychological Bulletin*, 76(2): 128–48.

Gelman, R. (1978), 'Cognitive Development', in L.W. Porter and M.R. Rosenweig (eds), *Annual Review of Psychology*, Palo Alto, CA: Annual Reviews, pp. 297–332.

Graesser, G.C. and N.H. Anderson. (1974), 'Cognitive Algebra of the Equation: Gift size = Generosity × Income', *Journal of Experimental Psychology*, 103(4): 692–9.

Gupta, M. and R. Singh. (1981), 'An Integration—Theoretical Analysis of Cultural and Developmental Differences in Attribution of Performance', *Developmental Psychology,* 17(6): 816–25.

Hau, K.T. and F. Salili. (1996), 'Prediction of Academic Performance Among Chinese Students: Effort can Compensate for Lack of Ability', *Organizational Behavior and Human Decision Processes*, 62(1): 3–94.

Heider, Fritz. (1958), *The Psychology of Interpersonal Relations*, New York: Wiley.

Herzberg, F.B., B. Mausner and B.B. Snyderman. (1959), *The Motivation to Work*, New York: Wiley.

Homans, George Caspar. (1961), *Social Behavior: Its Elementary Forms*, New York: Harcourt, Brace, and World.

Kun, A., J.E. Parsons and D.N. Ruble. (1974), 'Development of Integration Processes Using Ability and Effort Information to Predict Outcome', *Developmental Psychology*, 10(5): 721–32.

Lehman, D.R., C.Y. Chiu, and M. Schaller. (2004), 'Psychology and Culture', in Susan T. Fiske (ed.), *Annual Review of Psychology*, Palo Alto, CA: Annual Reviews, pp. 689–714.

Malpass, R.S. (1977), 'Theory and Method in Cross-Cultural Psychology', *American Psychologist*, 32(12): 1069–79.

Nebeker, D.M. (1975), 'Situational Favourability and Perceived Environmental Uncertainty: An Integrative Approach', *Administrative Science Quarterly*, 20(2): 281–94.

Riemer, B. (1975), 'Influence of Causal Beliefs on Affect and Expectancy', *Journal of Personality and Social Psychology*, 31(6): 1163–7.

Singh, R. 1975, 'Information Integration Theory Applied to Expected Job Attractiveness and Satisfaction', *Journal of Applied Psychology*, 60(5): 621–3.

_____. (1978), 'Disciplinary Judgments as Information Integration', *Journal of Social Psychology*, 104(2): 197–205.

_____. (1983), 'Leadership Style and Reward Allocation: Does Least Preferred Co-worker Scale Measure Task and Relation Orientation?', *Organizational Behavior and Human Performance*, 32(2): 178–97.

_____. (1985), 'A Test of the Relative Ratio Model of Reward Division with Students and Managers in India', *Genetic, Social, and General Psychology Monographs*, 111(3): 363–84.

_____. (1988), 'Life Performance ≠ Motivation × Ability × Opportunity: Individual Differences in Predictive Model', *Genetic, Social, and General Psychology Monographs*, 114(2): 191–210.

_____. (1991), 'Two Problems in Cognitive Algebra: Imputations and Averaging–versus—Multiplying', in Norman H. Anderson (ed.), *Contributions to Information Integration Theory, Vol. II: Social*, Hillsdale, NJ: Erlbaum, pp. 143–80.

Singh, R. and S. Bhargava. (1985), 'Motivation, Ability, and Exam Performance: Tests of Hypotheses of Cultural Difference and Task Difficulty', *Journal of Experimental Social Psychology*, 21(5): 466–79.

_____. (1986), 'Constant-Weight versus Relative-Weight Averaging in the Prediction of Exam Performance', *Journal of Experimental Social Psychology*, 22(6): 547–66.

Singh, R., K.A. Bohra, and A.K. Dalal. (1979), 'Favourableness of Leadership Situations Studied with Information Integration Theory', *European Journal of Social Psychology*, 9(3): 253–64.

Singh, R., M. Gupta, and A.K. Dalal. (1979), 'Cultural Difference in Attribution of Performance: An Integration–Theoretical Analysis', *Journal of Personality and Social Psychology*, 37(8): 1342–51.

Singh, R., U.R. Sidana, and S.K. Saluja. (1978), 'Playgroup Attractiveness Studied with Information Integration Theory', *Journal of Experimental Child Psychology*, 25(3): 429–36.

Singh, R. and P. Singh. (1994), 'Prediction of Performance Using Motivation and Ability Information: New Light on Integrational Capacity and Weighing Strategies', *Cognitive Development*, 9(4): 455–96.

Sinha, K.K. (1985), 'Developmental Trends in Distributive Justice: An Information Integration Analysis', Unpublished doctoral dissertation, University of Bihar, Muzaffarpur, India.

Srivastava, P. and R. Singh. (1988), 'Age and Task Differences in Prediction of Performance from Motivation and Ability Information', *Child Development*, 59(3): 769–81.

Surber, C.F. (1980), 'The Development of Reversible Operations in Judgments of Ability, Effort, and Performance', *Child Development*, 51(4): 1018–29.

_____. (1981a), 'Effects of Information Reliability in Predicting Task Performance Using Ability and Effort', *Journal of Personality and Social Psychology*, 40(6): 977–89.

_____. (1981b), 'Necessary versus Sufficient Causal Schemata: Attributions for Achievements in Difficult and Easy Tasks', *Journal of Experimental Social Psychology,* 17(6): 569–86.

_____. (1985), 'Measuring the Importance of Information in Judgment: Individual Differences in Weighting Ability and Effort', *Organizational Behavior and Human Decision Processes*, 35(2): 156–78.

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